Opinion Mining in Computational Social Science: A Systematic Review

Tables and Plots

Intro

Tables and plots to answer our research questions:

- RQ1a: Which theoretical concepts are being measured in opinion mining?
- RQ1b: Which methods and measurements are used for opinion mining?
- RQ1c: Which tools are used for opinion mining?
- RQ1d: Which data is used in CSS opinion mining?
- RQ2: Are there disciplinary differences in concepts, methods, tools, and data in opinion mining?
- RQ3: Do concepts, methods, tools, and data in opinion mining change over time?

Packages and functions

Data

Main

Validation

RQ1a theoretical concepts

Question Q4_1_Opinion-Evaluation

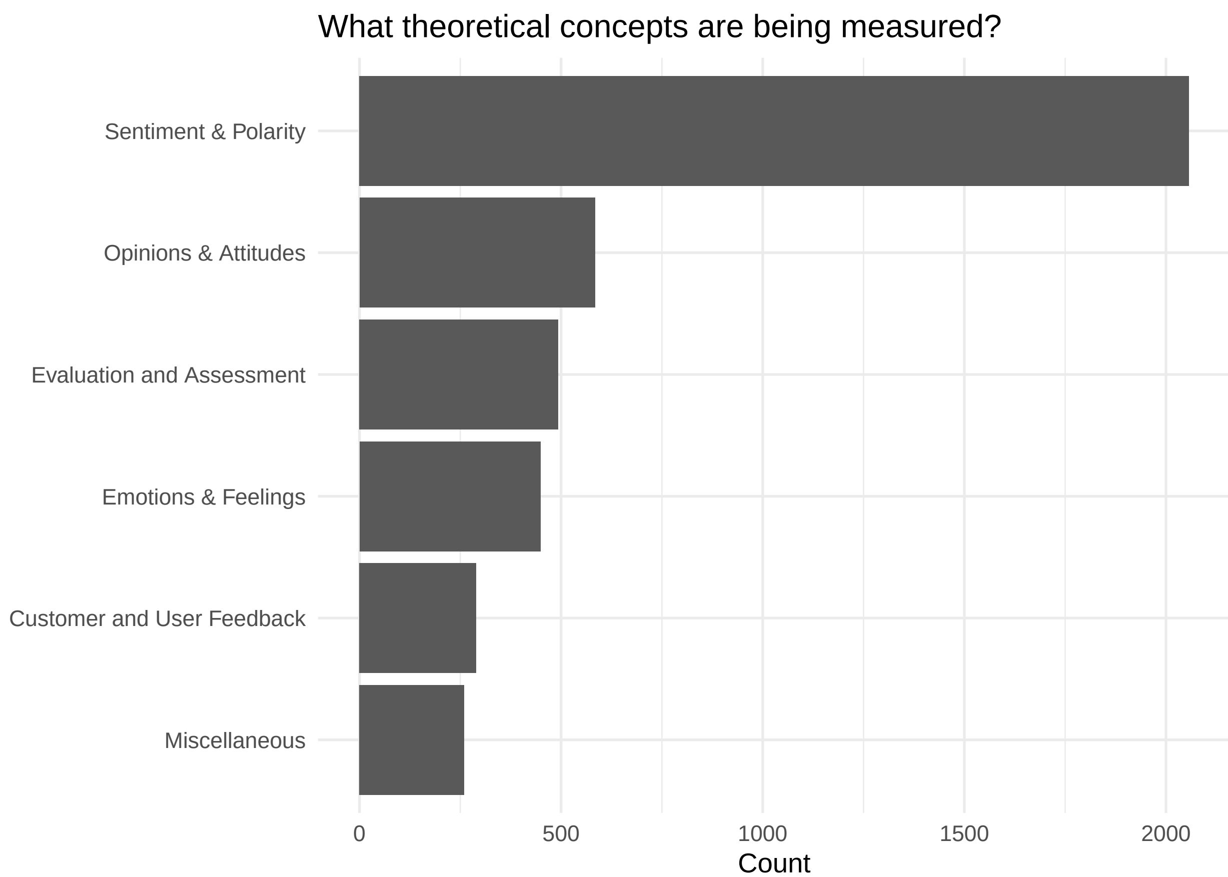


Figure 1: Categories of Opinion Evaluation Concepts

RQ1b methods and measurements used for opinion mining?

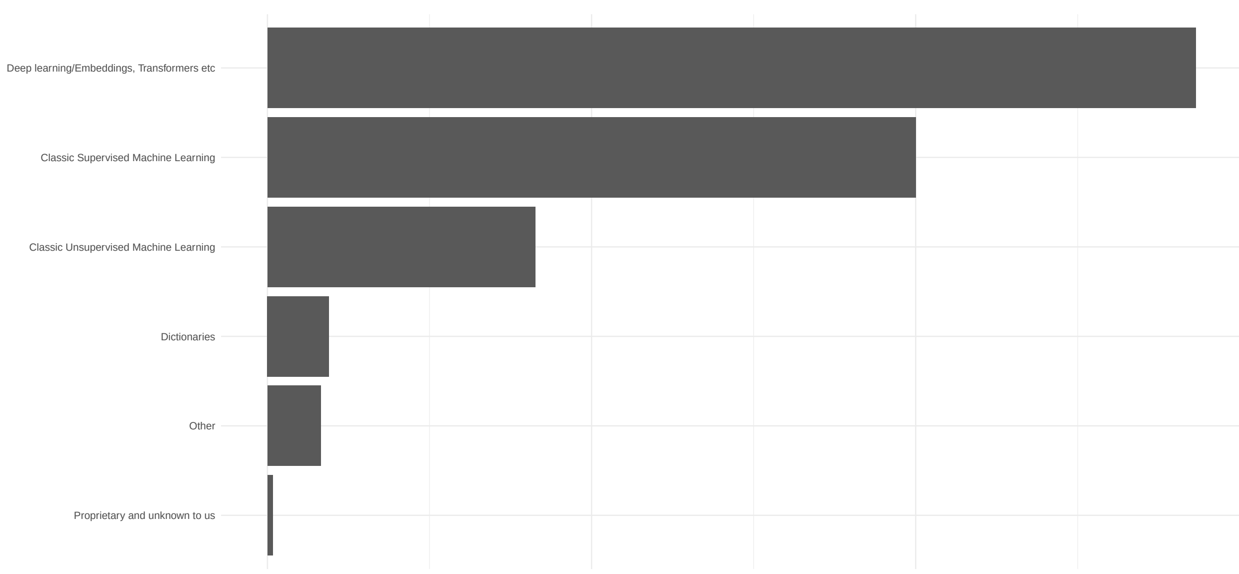


Figure 2: Approaches used for Opinion Measurement

RQ1c: Which tools are used for opinion mining?

Table 1: Most Frequently Used Tools for Opinion Mining

Tool	Category	Classification	Target	Software	N
SVM	Algorithm	x			189
Naive Bayes	Algorithm	x			171
VADER	Single Purpose Nlp	x		x	136
BERT	Model	x			131
LSTM	Algorithm	x			127
CNN	Algorithm	x			124
word2vec	Algorithm				105
TextBlob	General Purpose Nlp	x		x	99
Random Forest	Algorithm	x			78
LDA	Algorithm	x			77
SentiWordNet	Linguistic Resource	x			74
NLTK	General Purpose Nlp	x	x	x	72
Logistic Regression	Algorithm	x			66
TF-IDF	Algorithm				60
BiLSTM	Algorithm	x			56
GloVe	Algorithm				47
KNN	Algorithm	x			47
Decision Tree	Algorithm	x			40
Stanford CoreNLP	General Purpose Nlp	x	x	x	37
RNN	Algorithm				34
Multinomial Naive Bayes	Algorithm	x			31
SentiStrength	Single Purpose Nlp	x		x	30
WEKA	Gui Tool	x		x	29
fastText	General Purpose Nlp	x		x	29
spaCy	General Purpose Nlp	x	x	x	26

Note: N represents the number of unique papers that mention having used the tool in question for opinion mining (Q1.1). x indicate whether tool has been annotated as typically used for c

Table 2: Most Frequently Used Tools for Classification in Opinion Mining

Tool	Category	Software	N
SVM	Algorithm		189
Naive Bayes	Algorithm		171
VADER	Single Purpose Nlp	x	136
BERT	Model		131
LSTM	Algorithm		127
CNN	Algorithm		124
TextBlob	General Purpose Nlp	x	99
Random Forest	Algorithm		78
LDA	Algorithm		77
SentiWordNet	Linguistic Resource		74

Note: N represents the number of unique papers that mention having used the tool in question for opinion mining (Q1.1).

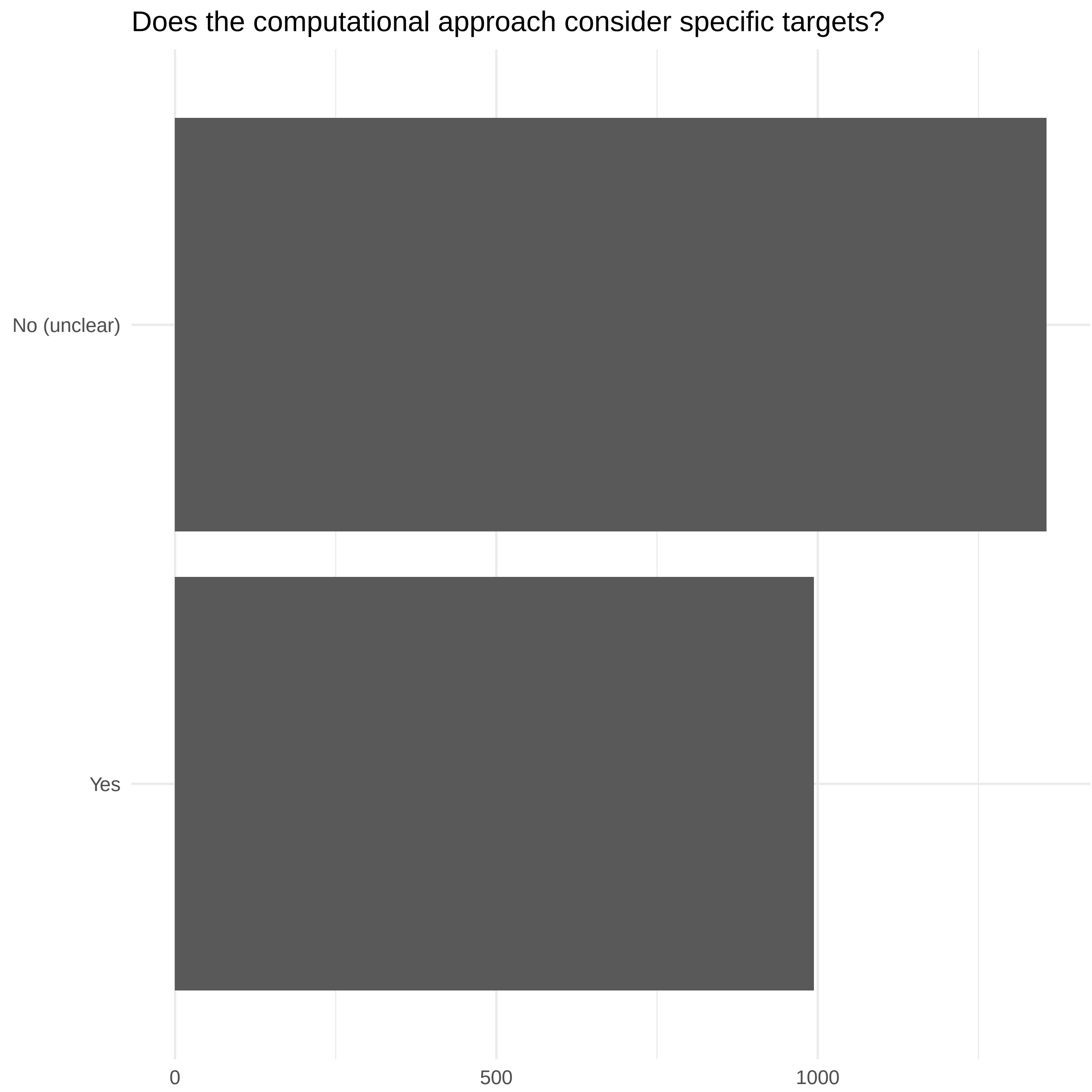
Table 3: Most Frequently Used Tools for Target-Identification in Opinion Mining

Tool	Category	Software	N
NLTK	General Purpose Nlp	x	72
Stanford CoreNLP	General Purpose Nlp	x	37
spaCy	General Purpose Nlp	x	26
RapidMiner	Gui Tool	x	20
CRF	Algorithm		19
Google Cloud Natural Language API	Commercial Api		8
MeaningCloud	Commercial Api		8
Apache OpenNLP	General Purpose Nlp	x	7
BiLSTM-CRF	Algorithm		6
OpinionFinder	Single Purpose Nlp	x	4

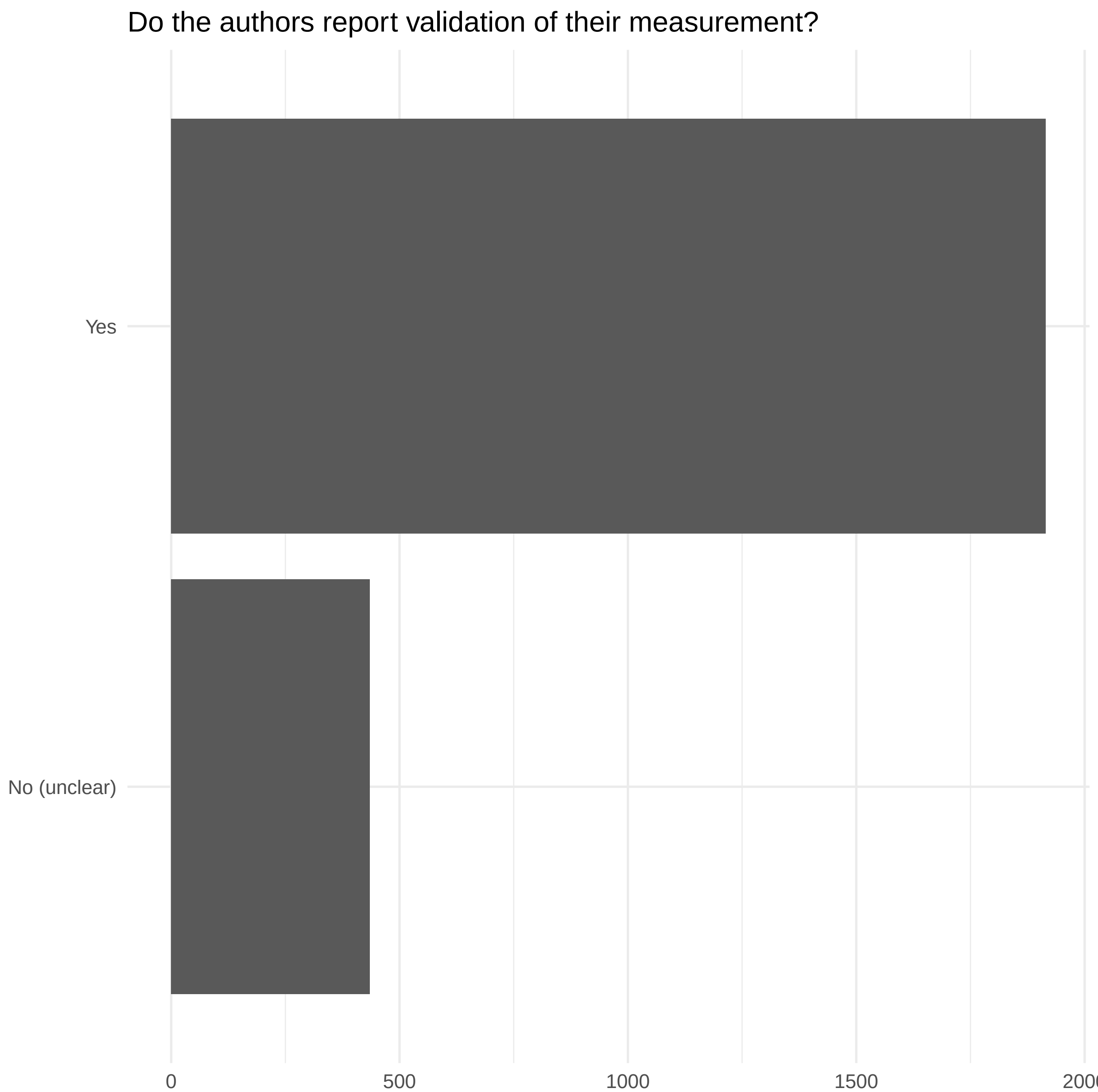
Note: N represents the number of unique papers that mention having used the tool in question for opinion mining (Q1.1).

RQ1d: Which data is used in CSS opinion mining?

Target



Validation


Do the authors report validation of their measurement?
Yes
No (unclear)
0
500
1000
1500
2000

RQ2: disciplinary differences

concepts

methods

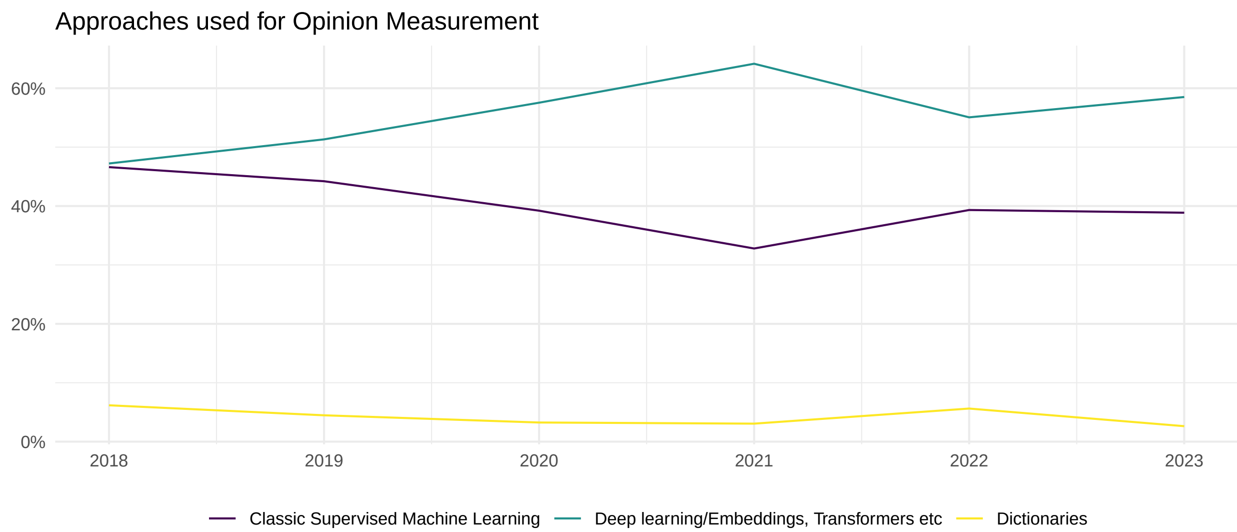
tools

data

RQ3: change over time

concepts

methods



tools

data

wrap up

Save data if it takes long to preprocess it here. Afterwards we get some information which is important to reproduce the report.

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Running under: Windows 11 x64 (build 26200)

Matrix products: default


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[3] LC_MONETARY=English_Europe.utf8 LC_NUMERIC=C
[5] LC_TIME=English_Europe.utf8
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attached base packages:
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other attached packages:
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 [9] ggplot2_4.0.1   tidyverse_2.0.0

loaded via a namespace (and not attached):
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